

Regularization

Regularizer

A **regularizer** is an **additional criterion to the loss function**

- It makes sure to **reduce the risk of overfitting**
- It tries to **keep the parameters more regular**
- → **It is a bias** on the model: forces to prefer certain types of weights

$$\operatorname{argmin}_{w,b} \sum_{i=1}^n L(y, y') + \lambda \cdot \Omega(w, b)$$

where L is the loss function,

Ω is the regularizer function and λ is a parameter associated with the regularizer

The case of LINEAR MODELS

In linear models the prediction function is:

$$\hat{y} = b + \sum_{j=1}^n w_j f_j$$

In this case we want:

- To **avoid huge weights**
(a small change in a feature can result in a large change in the prediction)
- **Prefer weights of 0** for **features** that **aren't useful**

p -norm

The p -norm measures the length of a vector in a p dimensional space:

- 1-norm: $r(w, b) = \sum |w_j|$
- 2-norm: $r(w, b) = \sqrt{\sum |w_j|^2}$
- p -norm: $r(w, b) = \sqrt[p]{\sum |w_j|^p} = \|w\|^p$



We can observe that:

- All p -norms **penalize larger weights**
- $p < 2 \rightarrow$ encourages sparser vectors
- $p \geq 2 \rightarrow$ discourages larger vectors

Gradient descent with a regularizer

To **apply gradient descent** we should **make sure** that the whole **objective function is a convex function**:

- That the **loss function** is a **convex** function
- That the **regularizer** is a **convex** function

We can **apply the gradient descent** and minimize the function:

- We **pick a starting point** w
- We **pick a dimension**
- We **move a small amount** in that dimension $\rightarrow$ **towards decreasing loss**
- **Repeat until loss doesn't decrease in any dimension**

$$w_j = w_j - \eta \frac{d}{dw_j} (L(y, y') + \Omega(w, b))$$

$$w_j = w_j - \eta \frac{d}{dw_j} (L(y, y')) - \eta \frac{d}{dw_j} (\Omega(w, b))$$

LP regularizer

From the p -norms we can define some regularizer with a derivative easier to compute:

- By elevating the p -norm by p we can remove the root
- By dividing λ by p , we define a new constant that makes the differentiation easier

L1 regularizer

To 1-norm we associate:

Regularizer: $\sum_j |w_j|$ **Constant:** λ

L2 regularizer

To 2-norm we associate:

Regularizer: $\sum_j w_j^2$ **Constant:** $\frac{\lambda}{2}$

LP regularizer

To p -norm we associate:

$$\text{Regularizer: } \sum_j |w_j|^p \quad \text{Constant: } \frac{\lambda}{p}$$

The case of LINEAR MODELS

For example observe that:

- The **exponential loss function** is a **convex** function
- The **p -norm function** is **convex** $\forall p \geq 1$

An example for linear models is:

$$\operatorname{argmin}_{\mathbf{w}, b} \sum_{i=1}^n \exp(-y_i(\mathbf{w} \cdot \mathbf{x}_i + b)) + \frac{\lambda}{2} \sum_j |w_j|^2$$

In this case **we apply the 2-norm function as regularizer**
(and divide λ by 2 to make the differentiable computation easier)

L1 regularization

$$\text{Objective function: } \operatorname{argmin}_{\mathbf{w}, b} \sum_{i=1}^n \exp(-y_i(\mathbf{w} \cdot \mathbf{x}_i + b)) + \lambda \sum_j |w_j|$$

$$\text{Single update: } w_j \leftarrow w_j + \eta (\text{gradient loss} - \lambda \operatorname{sign}(w_j))$$

L2 regularization

$$\text{Objective function: } \operatorname{argmin}_{\mathbf{w}, b} \sum_{i=1}^n \exp(-y_i(\mathbf{w} \cdot \mathbf{x}_i + b)) + \frac{\lambda}{2} \sum_j |w_j|^2$$

$$\text{Single update: } w_j \leftarrow w_j + \eta (\text{gradient loss} - \lambda w_j)$$

LP regularization

$$\text{Objective function: } \operatorname{argmin}_{\mathbf{w}, b} \sum_{i=1}^n \exp(-y_i(\mathbf{w} \cdot \mathbf{x}_i + b)) + \frac{\lambda}{p} \sum_j |w_j|^p$$

$$\text{Single update: } w_j \leftarrow w_j + \eta (\text{gradient loss} - \lambda w_j^{p-1})$$

However, $\frac{\lambda}{p}$ can be approximated to λ

In general we have that:

- If w_j is **positive**, the **regularizer is subtracted** $w_j \rightarrow \text{decreases}$
- If w_j is **negative**, the **regularizer is added** $w_j \rightarrow \text{increases}$

Concerning the p -norm regularization:

- **Without magnitude in L1 regularization** (we consider sign only)
- **With growing magnitude as p increases**